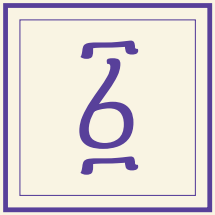
# Business cases

*The inner product and nearest neighbors*

Carlos Cotrini
*AI in Industry, ETH Zurich*
*Friday 13 November 2026* 



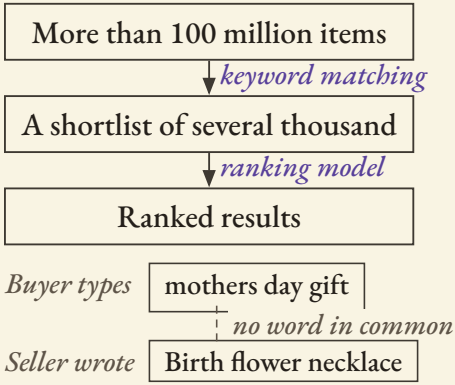
*Section 1*

# Etsy

# Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.




*Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.*

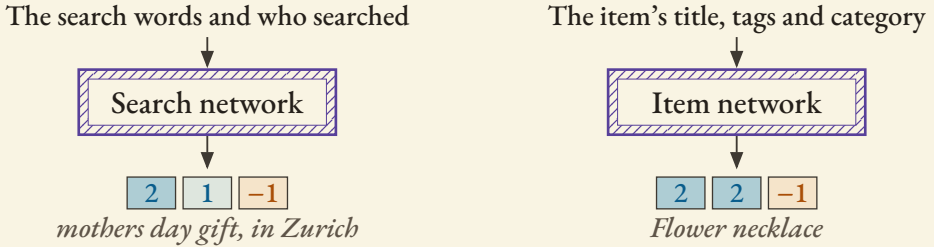
# The training data: a month of search logs

1 month: more than 150 million rows, from up to 30 million users.

| Search | Who searched | Item | Action |
|---|---|---|---|
| mothers day gift | Zurich, searched necklaces | Flower necklace | purchase |
| bridge photos | New York, clicked print shops | Bridge print | cart add |
| wool socks | Oslo, searched knitting | Knitted socks | cart add |

*The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.*

# Every search and every item becomes a list of numbers




Training makes the inner product high for items bought or added to the cart after the search: here 7.

*Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.*

# A search arrives: sort by inner product

| | | | | inner product |
|---|---|---|---|---|
| *mothers day gift* | 2 | 1 | −1 | |
| | × | × | × | |
| Flower necklace | 2 | 2 | −1 | |
| | 4 | + 2 | + 1 | = 7 |
| Gift wrap | −1 | 0 | 2 | −4 |
| Flower necklace | 2 | 2 | −1 | 7 |
| Photo mug | 1 | 1 | 0 | 3 |
| Recipe board | 1 | 2 | 0 | 4 |

*Invented numbers.*

*Sorted: the top two join the shortlist*

| | |
|---|---|
| Flower necklace | 7 |
| Recipe board | 4 |
| Photo mug | 3 |
| Gift wrap | −4 |

# Cost and value

*Cost*

**72 hours** *on 4 GPUs*

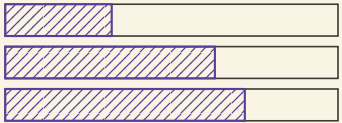
**1 month** *of search logs*

*Value: A/B tests, relative change*

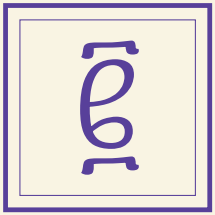
**+2.63%** *visits that end in a purchase*

**+5.58%** *searches that end in a purchase*

*Popular searches: share of the bought products found in the top 100*




*Jha and others (Etsy), IEEE FMLDS 2024.*



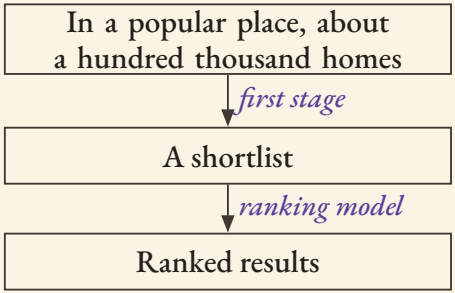
*Section 2*

# Airbnb

# Airbnb: the first stage of search

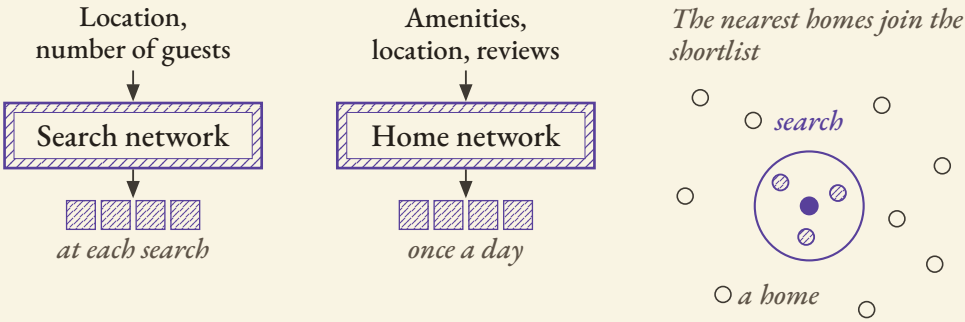
The first stage evaluates tens of thousands of homes.

It used to be only a simple linear model that scored every home once a day, whatever the search.



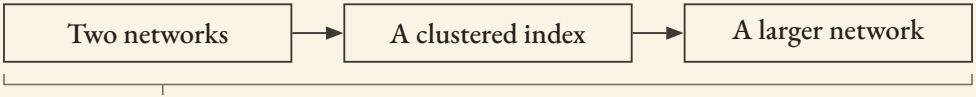

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# Two networks and the nearest homes




*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.*

# The value: three A/B tests




**+0.31%**

*visits that end in a booking, all three, relative change*

Airbnb calls this gain “a large portion of the team’s annual goal”.

**+2.3%** *bookings from promotional emails*

**−16%** *compute with the clustered index*

*Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.*